

Exam: ETLD
Duration: 1h 30'

Module: ASD1
LI – S1 – 2024/2025

Exercise 1 : (5 pts)

Given the following character strings:

```
char str1[ ] = "exam";  
char str2[ ] = "ASD1";  
char str3[15];
```

Write the necessary instructions in **C language** to perform the following operations **without using loops**:

1. Copy the content of str1 into the string str3[].
2. Add the string str2 to str3[], inserting one space between the two words.
3. Replace the first character of str3 with the character 'E'.
4. Display str3.

Exercise 2 : (6 pts)

Given the following series:

$$S(x, n) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \dots + \frac{x^n}{n!}$$

Write an algorithm or a C program that calculates and displays the sum $S(x, n)$ for given non-null values of x and n.

Remark: Do not use predefined functions to calculate powers or factorials.

Exercise 3 : (9 pts)

Let A be an array of integers. Write an algorithm or C program that:

- 1- Fill the array A with N integers such that $1 \leq A[i] \leq 10$ for all i.
- 2- Calculate the number of occurrences (number of repetitions) of each element of A and store the occurrences in the array T.
- 3- Create a matrix M (N x N) from A satisfying the following conditions:
 - a. The matrix M should contain **only** the elements of the array A.
 - b. Each row in M contains **distinct** integers (without repetition).
 - c. The number of rows in M should be **minimal**.
 - d. The cells that don't contain elements of the matrix M are filled with 0.
- 4- Display the non-null values of M.

Example : N=7

A =	1	3	4	1	2	3	1
	0	1	2	3	4	5	6

M is displayed:

```
1 3 4 2  
1 3  
1
```

M=

	0	1	2	3	4	5	6
0	1	3	4	2	0	0	0
1	1	3	0	0	0	0	0
2	1	0	0	0	0	0	0
3	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0

Good luck